L1.8 Linearization & Taylor

§2.9, §11.10 — what you need to know

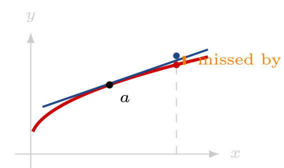
LINEARIZATION

$$\boxed{L(x) = f(a) + f'(a)(x-a)}$$

"the tangent line, used as a calculator"

Two ingredients, both evaluated **at** a: the height and the slope. Choose a close to the value you want, and somewhere you can evaluate f exactly.

Small angle. At $a = 0$: $\sin x \approx x$, $\cos x \approx 1$, $\tan x \approx x$ for $x \lesssim 10°$. Radians only — it sets an angle equal to a length.



ACTUAL CHANGE AND THE DIFFERENTIAL

$$\boxed{\Delta y = f(x+\Delta x) - f(x) \qquad dy = f'(x)\,dx}$$

"Δy is what happened; dy is what the line predicted"
Take $dx = \Delta x$ so both describe the same step in. The gap between them is exactly **how far the tangent line's estimate missed**, and it closes far faster than the step does — halve dx and the gap drops by about four.

FRACTIONAL UNCERTAINTY

A measurement arrives as a value and an **uncertainty**, $r = 21 \pm 0.05$ cm. That uncertainty **is** a dr, and anything computed from it inherits one: $dV = f'(r)\,dr$. *Not the same thing as the gap above* — that was how far an estimate missed; this is how well a measurement is known.

$$\boxed{\frac{df}{f} = n\left(\frac{dx}{x}\right) \quad \text{for} \quad f = Cx^n}$$

"the exponent becomes the multiplier"
A 1% uncertainty in a radius is 3% in a volume, 2% in an area, 0.5% in a pendulum's period.

Report both — the absolute uncertainty carries units, the fractional one transfers: $V = 38{,}792 \pm 277$ cm^3, $\dfrac{dV}{V} = 0.71\%$.

SUMMATION CLOSED FORMS

$\sum_{i=1}^{n} a_i = a_1 + a_2 + \cdots + a_n$, where i is the **index**. Given, not derived — and the third is the square of the first.

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2} \qquad \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6} \qquad \sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$$

TAYLOR AND MACLAURIN SERIES

$$\boxed{f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n \qquad a = 0 \;\Rightarrow\; \text{Maclaurin}}$$

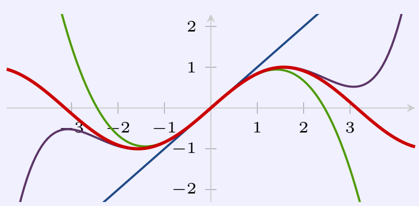
"differentiate n times, evaluate at a, divide by $n!$"

a is **where you aim it** — same function, different centre, different series, each best near its own. The three worth knowing cold, all at $a = 0$:

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots \qquad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \qquad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

The truncation ladder

One formula, cut off at different places. Drawn on $\sin x$ about $a = 0$, whose even terms are all zero.



$\sin x$	the function
$T_1 = x$	**this is** $L(x)$
$T_3 = x - \frac{x^3}{3!}$	
$T_5 = x - \frac{x^3}{3!} + \frac{x^5}{5!}$	

All of them are exact at a and get worse as you walk away. More terms buys *more room*, not accuracy everywhere.

Keep the $n = 0$ and $n = 1$ terms of the Taylor series and you get $f(a) + f'(a)(x-a)$. **Linearization is the degree-1 Taylor polynomial** — the two halves of this lesson are one idea at two truncations.

WATCH OUT

✗ ~~linearization is a new idea~~ → **it is the tangent line you already had, given a new job**

✗ ~~Δy and dy are the same~~ → Δy **happened,** dy **was predicted — the gap is how far the estimate missed**

✗ ~~the uncertainty in V is the uncertainty in r~~ → **it is three times the fractional uncertainty, since** $V \propto r^3$

✗ ~~$\sin x \approx x$ always~~ → **only for small** x**, and only in radians**

✗ ~~a Maclaurin series works everywhere~~ → **it is built at** $a = 0$ **and is best *near*** $a = 0$

✗ ~~more terms is always better~~ → **it helps near** a**; far from** a **it can still be useless**